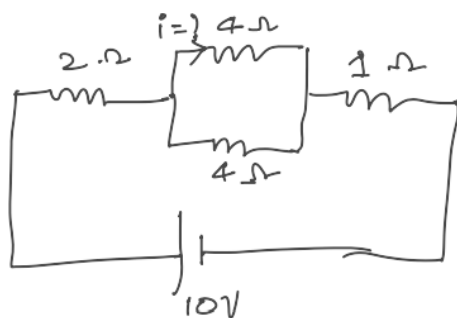


**JEE-Main-29-01-2024 (Memory Based)**  
**[EVENING SHIFT]**

**Physics**

**Question:** Find out the current  $i$ .



**Options:**

- (a) 1 A
- (b) 2 A
- (c) 3 A
- (d) 4 A

**Answer: (a)**

**Question:** Two equal charges of masses  $m_1$  &  $m_2$  are sent in a transverse magnetic field by accelerating through same potential difference. Find the ratio of their radii inside?

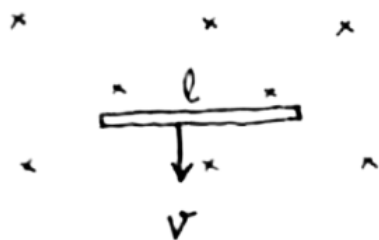
**Options:**

- (a)  $\sqrt{\frac{m_2}{m_1}}$
- (b)  $\sqrt{\frac{m_1}{m_2}}$
- (c)  $\frac{m_1}{m_2}$
- (d)  $\frac{m_2}{m_1}$

**Answer: (b)**

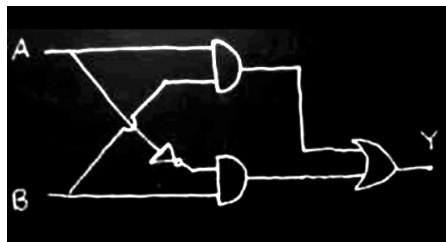
**Question:** A rod is dropped as shown where horizontal component of Earth's magnetic field is  $B_H$ .

Find EMF (t)?



$$B_H = 0.6 \times 10^{-4} \text{ T}$$

$$V = 10 \text{ m/s } l = 5 \text{ m}$$

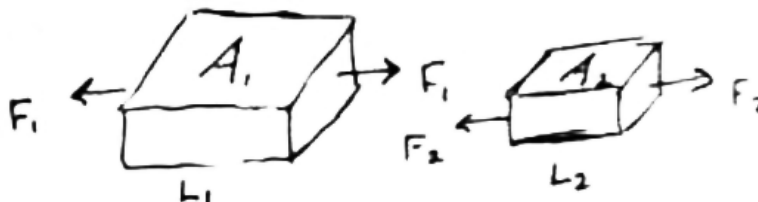


Question:

Find Y truth table?

Question: Two blocks of equal volume have same elongations for deforming forces find  $F_1/F_2$ ?

$$\frac{A_1}{A_2} = \frac{4}{1}$$

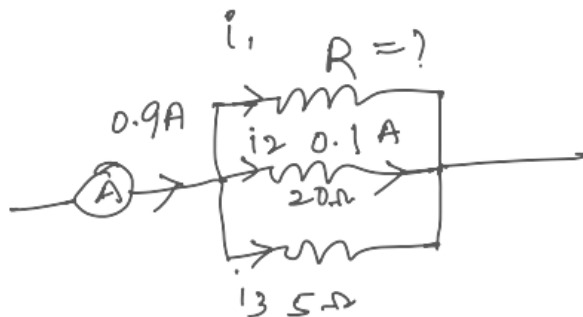


Options:

- (a) 4 : 1
- (b) 1 : 4
- (c) 16 : 1
- (d) 1 : 16

Answer: (c)

Question: Find out The resistance R in the Given Circuit



Options:

- (a) 2  $\Omega$
- (b) 3  $\Omega$
- (c) 4  $\Omega$
- (d) 5  $\Omega$

Answer: (d)

**Question:** Time period of a particle performing SHM is  $6\pi$  s. Find the time taken by the particle to go from  $x = A$  to  $x = A/2$

**Options:**

- (a)  $\pi$  s
- (b)  $\pi/2$  s
- (c)  $3\pi/2$  s
- (d)  $3\pi$  s

**Answer: (a)**

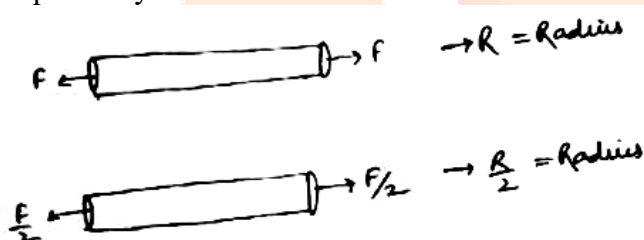
**Question:** For an ideal gas, pressure is 1.38 atm and number of molecules are  $2 \times 10^{25}$  per  $\text{m}^3$ . Find the temperature of the gas ?

**Options:**

- (a) 1500 K
- (b) 1000 K
- (c) 500 K
- (d) 250 K

**Answer: (c)**

**Question:** Two Rods of same length and material is applied with the forces  $F$  and  $F/2$  respectively. If the cross sectional radii are  $R$  and  $R/2$  then find the ratio of the extensions



**Options:**

- (a) 2:1
- (b) 1:2
- (c) 1:4
- (d) 4:1

**Answer: (b)**

**Question:** A particle is tied to a rope. If its moving such that it just completes the vertical circle. Find the ratio of kinetic energy at lowermost point & upper most point respectively?

**Options:**

- (a) 5:1
- (b) 3:2
- (c) 2:1
- (d) 1:5

**Answer: (a)**

**Question:** Bob of pendulum of length  $l$  is released from horizontal position. If 10% of energy is lost then find the velocity on reaching the lowest point

**Options:**

- (a)  $\sqrt{(9gl/5)}$
- (b)  $\sqrt{(3gl/5)}$
- (c)  $\sqrt{(3gl)}$
- (d)  $\sqrt{(5gl)}$

**Answer: (a)**

**Question:** Two particles each of charge  $q$  are accelerated by same potential difference and projected into the same magnetic field. Ratio of Radii is given then find the ratio of Mass.

**Question:** A planet revolving around sun in a circular orbit of radius  $R$  has a time period  $T_1$ . Another planet revolving around sun in a circular orbit of radius  $R/4$  has a time period  $T_2$ . Find  $T_2/T_1$

**Options:**

- (a) 1:8
- (b) 8:1
- (c) 4:1
- (d) 1:4

**Answer: (a)**

**Question:** S1: Positive charge is present on the nucleus and electrons revolve around the nucleus in Rutherford's model

S2: Plum pudding is a special case of Rutherford model.

**Options:**

- (a) Both S1 & S2 are false
- (b) Both S1 & S2 are true
- (c) S1 is true but S2 is false
- (d) S2 is true but S1 is false

**Answer: (c)**

**Question:** In a single slit diffraction experiment, wavelength used is  $6000 \text{ \AA}$ . The distance between 1st and 3rd minima is 3mm. If screen is 50 cm away from the slit, find the slit width

**Options:**

- (a) 0.1 mm
- (b) 0.2 mm
- (c) 0.3 mm
- (d) 0.4 mm

**Answer: (b)**

**Question:** An electromagnetic wave is travelling in positive  $x$  direction. Electric field at a location is given by  $\vec{E} = 9.6\hat{j} \text{ (N/C)}$ . What is the value of  $\vec{B}$  at this location?

**Options:**

- (a)  $3.2 \times 10^{-8} \hat{k} \text{ Tesla}$
- (b)  $28.8 \times 10^8 \hat{k} \text{ Tesla}$
- (c)  $-3.2 \times 10^{-8} \hat{k} \text{ Tesla}$
- (d)  $-28.8 \times 10^8 \hat{k} \text{ Tesla}$

**Answer: (a)**

**Question:** The distance between real object and virtual Image is given that is 15 cm, the magnification is 2. Find the focal length of Mirror.

**Options:**

- (a)  $f = -10 \text{ cm}$
- (b)  $f = 10 \text{ cm}$
- (c)  $f = -5 \text{ cm}$

(d)  $f = 5$  cm

**Answer: (a)**

**Question:**  $n$  moles of triatomic gas ( $f = 6$ ) and 2 mole of monatomic gas are mixed together to give a mixture of 5 degrees of freedom. Find ' $n$ '

**Options:**

(a) 1

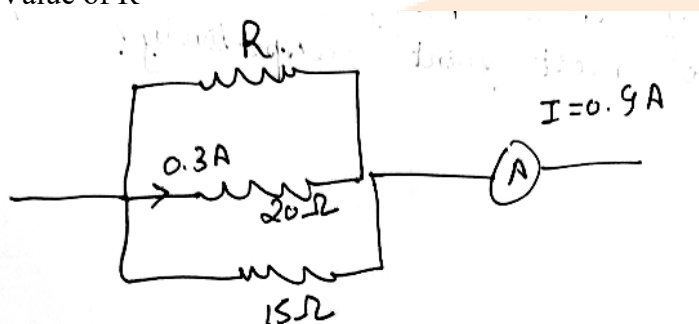
(b) 4

(c) 3

(d) 5

**Answer: (b)**

**Question:** Below Current is given & Ammeter Reads 0.9 A & Current in  $20\ \Omega$  is 0.3 A. Find Value of R



**Options:**

(a)  $10\ \Omega$

(b)  $20\ \Omega$

(c)  $30\ \Omega$

(d)  $40\ \Omega$

**Answer: (c)**

**Question:** Electric field is given in a region  $\vec{E} = 6\hat{i} + 5\hat{j} + 3\hat{k}$  N/C. Find flux linkage through a surface area  $30\text{ m}^2$  that is in YZ plane?

**Options:**

(a) 100 Wb

(b) 130 Wb

(c) 150 Wb

(d) 180 Wb

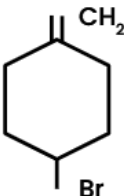
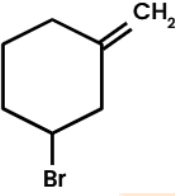
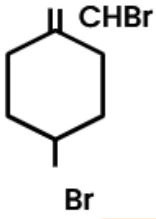
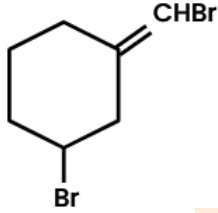
**Answer: (d)**

**JEE-Main-29-01-2024 (Memory Based)**  
**[EVENING SHIFT]**

**Chemistry**

**Question:** Which can show GI.

**Options:**

- (a) 
- (b) 
- (c) 
- (d) 

**Answer: (d)**

**Solution:**

**Question:** What is the oxidation number of iron in the compound formed in brown ring test of  $\text{NO}_3^-$

**Options:**

- (a) +1  
(b) +2  
(c) +3  
(d) -1

**Answer: (a)**

**Solution:**

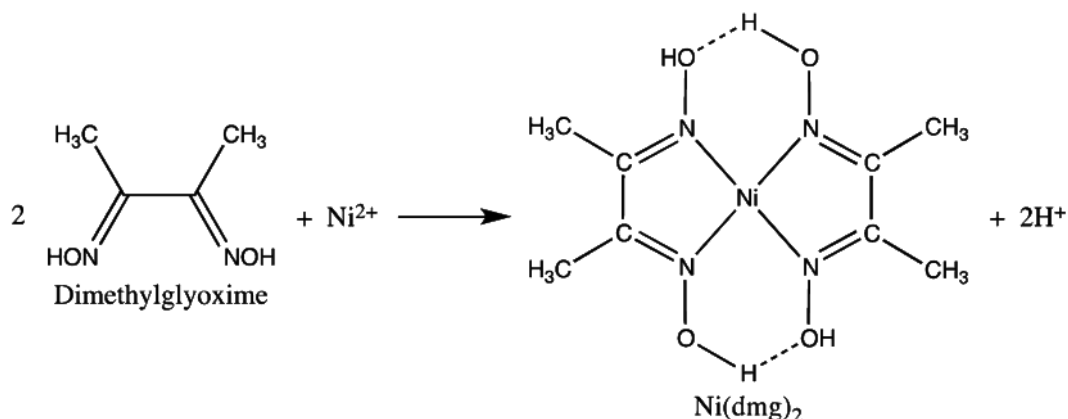
+1 oxidation state of Fe

The brown ring complex compound is formulated as  $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ .

**Question:** Which reagent is used for getting red colour with nickel ions  $\text{Ni}^{+2}$

**Options:**

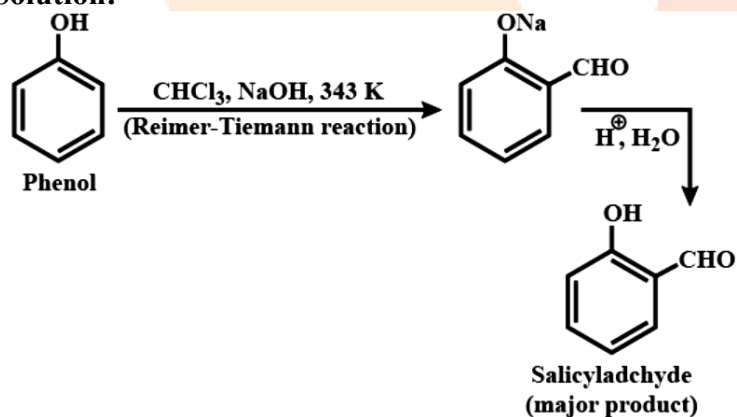
- (a) EDTA
- (b) Dimethylglyoxime
- (c)  $\alpha$  - nitroso -  $\beta$  - naphthol
- (d) None of the above

**Answer: (b)****Solution:**

**Question:** Phenol is reacted with chloroform in the presence of NaOH and the obtained product is hydrolyzed in presence of acid. The final product is

**Options:**

- (a) Benzene-1,2-diol
- (b) Benzene-1,3-diol
- (c) Salicylaldehyde
- (d) Hydroxybenzaldehyde

**Answer: (c)****Solution:**

**Question:** Which ion gives brownish colour with nessler's reagent

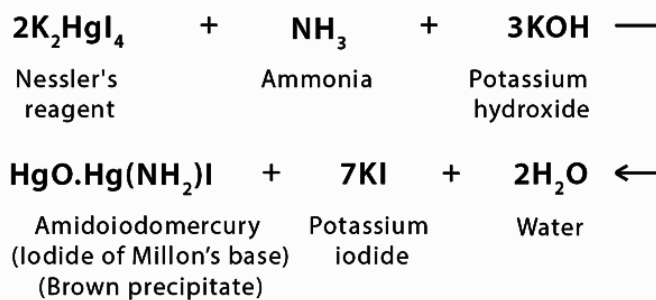
**Options:**

- (a) Sulphate salt
- (b) Nitrate ion
- (c) Bromide ion
- (d) Ammonium salt

**Answer: (d)**

**Solution:**

## Nessler's Reagent Reaction



**Question:** Arrange the following compounds according to the pKa value.

- Phenol
- Meta nitrophenol
- Para nitrophenol
- Ethanol

**Options:**

- $d > a > b > c$
- $a > b > c > d$
- $b > c > d > a$
- $c > d > b > a$

**Answer: (a)**

**Solution:**

**Question:** Best reducing agent among the given ions are :

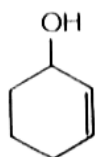
**Options:**

- $\text{Ce}^{4+}$
- $\text{Gd}^{2+}$
- $\text{Lu}^{3+}$
- $\text{Nd}^{3+}$

**Answer:**

**Solution:**

**Question:** IUPAC Name of the compound is

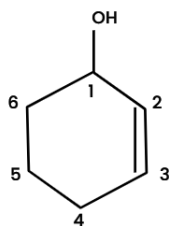


**Options:**

- Hex - 2 - en - 1 - ol
- Cyclohex - 2 - en - 1 - ol
- 3 - Hydroxycyclohexane
- Cyclohex - 1 - en - 3 - ol

**Answer: (b)**

**Solution:**



Cyclohex - 2 - en - 1 - ol

**Question:** Why does oxygen show anomalous behaviour in group 16 in the periodic table?

**Options:**

- (a) Large size, high electronegativity
- (b) Small size, small electronegativity
- (c) Small size, high electronegativity, absence of vacant d - orbit
- (d) Large size high electronegativity presence of vacant

**Answer: (c)**

**Question:** How many of the following compounds have zero dipole moment.

$\text{NH}_3$  ,  $\text{H}_2\text{O}$  ,  $\text{HF}$ ,  $\text{CO}_2$  ,  $\text{SO}_2$  ,  $\text{BF}_3$  ,  $\text{CH}_4$

**Solution:**

$\text{NH}_3 \Rightarrow$  Dipole Moment  $\neq 0$

$\text{BF}_3 \Rightarrow$  Dipole Moment  $= 0$

$\text{SO}_2 \Rightarrow$  Dipole Moment  $\neq 0$

$\text{CO}_2 \Rightarrow$  Dipole Moment  $= 0$

$\text{CH}_4 \Rightarrow$  Dipole Moment  $= 0$

$\text{HF} \Rightarrow$  Dipole Moment  $\neq 0$

$\text{CO}_2$ ,  $\text{BF}_3$  and  $\text{CH}_4$  Dipole Moment is zero

**Question:** Statement 1( $S_1$ ): F has highest EGE in its grp

Statement 2( $S_2$ ) : O has 2<sup>nd</sup> most EGE in its grp.

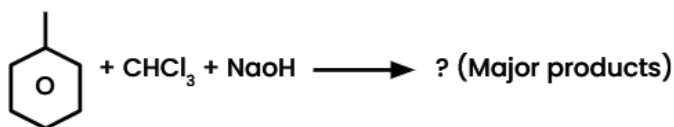
**Options:**

- (a) Both statement I and statement II are false
- (b) Statement I is true but statement II is false
- (c) Statement I is false but statement II is true
- (d) Both statement I and statement II are true

**Answer: (c)**

**Solution:**

1<sup>st</sup> statement wrong 2<sup>nd</sup> statement right



**Question:**

**Question:**  $S_1$  : Rutherford said mass is at the centre and charge is distributed.

$S_2$  :  $e^-$  are clustered

**Question:** Number of antibonding in 1s & 2p of diatomic  $\alpha$  diatomic molecule ?

**Question:** Find total number of  $\pi$  &  $\sigma$  bond in 2 formyl hex - 4 - enoic acid

**Options:**

(a) Sigma bonds = 18

Pi bonds = 3

(b) Sigma bonds = 16

Pi bonds = 2

(c) Sigma bonds = 16

Pi bonds = 3

(d) Sigma bonds = 18

Pi bonds = 2

**Answer: (a)**

**Solution:**

2 - Formyl hex - 4 - enoic acid  $(\text{CH}_3\text{-CH=CH-CH}_2\text{-CH(CHO)-COOH})$

$\sigma$  bonds - 18

$\pi$  bonds - 3

**Question:** Radio activity

Half life was given 36 hours.

Find amount left after 1 day log value was given.

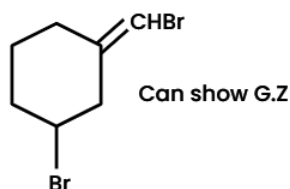
**Solution:**

$$K = \frac{0.693}{36} \text{ hrs}^{-1}$$

$$\frac{0.693}{36} \times 34 = 2.303 \log \frac{1}{A_k}$$

**Question:** IUPAC name  $\text{K}_2 [\text{MnO}_4]$  according to coordination compound

**Solution:**



**Question:** Which of them is a strong RA

**Options:**

(a)  $\text{Ce}^{4+}$

(b)  $\text{Ga}^{3+}$

(c)  $\text{Tb}^{3+}$

(d)  $\text{Ho}^{2+}$

**Answer:**

**Solution:**

**Question:** Number of antibonding in 1s & 2p of diatomic  $\alpha$  diatomic molecule ?

**Question:** Find molality of 0.8M  $\text{H}_2\text{SO}_4$  (density= $1.06\text{g/cm}^3$ ) we have to give answer in  $\underline{\hspace{1cm}} 10^{-3}$

**Solution:**

$$m = \frac{1000 M}{M \times H_b - 1000 d} = \frac{1000 \times 0.8}{0.8 \times 98.10^3 \times 1.06}$$

**Question:** Which of the following element has highest 1<sup>st</sup> ionization energy ?

**Options:**

- (a) N
- (b) C
- (c) Si
- (d) Al

**Answer: (a)**

**Question:** Match the following :

- |             |               |
|-------------|---------------|
| (A) Lyman   | (I) IR        |
| (B) Balmer  | (II) IR       |
| (C) Paschen | (III) Visible |
| (D) p-fund  | (IV) UV       |

**Solution:**

A-U<sub>v</sub>

B-Visible

C-IR

D-IR

**Question:** If standard enthalpy of vaporization of CCl<sub>4</sub> is 30.5 kJ/mol, find heat absorbed for vaporization of 294 gm of CCl<sub>4</sub>. [Nearest integer ] [in kJ/mol]

**Solution:**

1 mole → 30.5 kJ/mol

154 gm → 30.5 kJ/mol

$\Delta H_{\text{vap}}$

294 gm

mol 1.g → ?  $\Delta H_{\text{vap}} = 58.2$

**Question:** 50 mL of 0.5 oxalic acid is completely neutralized by 25 mL of NaOH solution. Find out the amount of NaOH (in gm) present in 25 mL of given NaOH solution.

**Question:** Match the following

- |                   |                                        |
|-------------------|----------------------------------------|
| (A) Starch        | (I) Peptide linkage                    |
| (B) Cellulose     | (II) $\alpha$ -D-Glycosidic linkage    |
| (C) Proteins      | (III) $\beta$ - D - Glycosidic linkage |
| (D) Nucleic acids | (IV) nucleotide                        |

**Solution:**

Starch - alpha d glucose

Cellulose - beta d glucose

Protein - peptide

Nucleic acid - nucleotide

# JEE-Main-29-01-2024 (Memory Based)

## [EVENING SHIFT]

### Maths

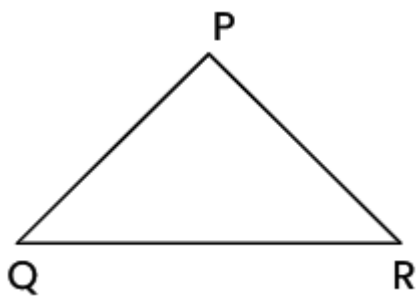
**Question:**  $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1 - \sin 2x} \, dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3}$

Find  $\alpha + 3\beta$ ; find  $\alpha\beta\gamma$

**Solution:**

$$\begin{aligned} & \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1 - \sin 2x} \, dx \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} |\cos x - \sin x| \, dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} |(\sin x - \cos x)| \, dx \\ &= +\sin x + \cos x \Big|_{\frac{\pi}{6}}^{\frac{\pi}{4}} + -\cos x - \sin x \Big|_{\frac{\pi}{6}}^{\frac{\pi}{4}} \\ &= \left( \frac{1}{\sqrt{2}} - \frac{1}{2} + \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \right) + \left( \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} - \frac{1}{2} \right) \\ &= 2\sqrt{2} - \sqrt{3} - 1 \end{aligned}$$

**Question:** If  $P(3, 2, 3)$   $Q(4, 6, 2)$   $R(7, 3, 2)$  are the vertices of  $\Delta PQR$ , then find  $\angle QPR =$



**Options:**

- (a)  $\cos^{-1} \frac{1}{18}$
- (b)  $\frac{\pi}{6}$
- (c)  $\frac{\pi}{3}$
- (d)  $\cos^{-1} \frac{7}{18}$

**Solution:**

$$\overline{PQ} = (1, 4, -1)$$

$$\overline{PR} = (4, 1, -1)$$

$$\cos x = \frac{4 + 4 + 1}{18}$$

$$x = \frac{\pi}{3}$$

**Question:** Area bounded by  $0 \leq y \leq \min\{x^2 + 2, 2x + 2\}$ ,  $x \in [0, 3]$ , then  $12A$  is

**Solution:**

$$\int_0^2 x^2 + 2 + \int_2^3 2x + 2 \frac{x^3}{3} + 2x$$

$$\left| 0^2 + x^2 + 2x \right| 2^3 \frac{8}{3} + 4 + 9 + 6 - (4 + 4)$$

$$A = \frac{41}{2}$$

$$12A = 164$$

**Question:** The remainder when  $64^{32^{32}}$  is divided by 9 is \_\_\_\_.

**Solution:**

$$64^{32^{32}} \rightarrow 9$$

$$64 = 1 \pmod{9}$$

$$64^{32} = 1 \pmod{9}$$

$$64^{32^{32}} = 1 \pmod{9}$$

**Question:**  $\cos(2 \sin^{-1} x) = \frac{1}{9}$  holds for  $x = \frac{m}{n}$  and  $\alpha\beta$  are the roots of the equation  $mx^2 - nx -$

$m + n = 0 (\alpha > \beta)$  then  $\alpha, \beta$  lies on the line

**Options:**

(a)  $5x - 8y = 9$

(b)  $5x + 8y = 9$

(c)  $8x + 5y = 9$

(d)  $8x - 5y = 9$

**Solution:**

$$\cos(\sin^{-1} x) = \frac{1}{9}$$

$$1 - 2\sin^2(\sin^{-1} x) = \frac{1}{9}$$

$$2x^2 = \frac{8}{9} \rightarrow x = \pm \frac{2}{3} = -\frac{2}{3}$$

$$\frac{2}{3} : 2x^2 - 3x + 1 = 9$$

$$\alpha = 1, \beta = \frac{1}{2}$$

$$5x + 8y = 9$$

**Question:** If  $\alpha$  &  $\beta$  are roots of the equation  $x^2 - \sqrt{6}x + 3$  and where  $\text{Im}(\beta) < 0$  if

$$\frac{\alpha^{99}}{\beta} + \beta^{98} = 3^n (a + ib) \text{ then find a, b, n.}$$

**Solution:**

$$x^2 - \sqrt{6}x + 3 = 0$$

$$\beta, \alpha = \frac{\sqrt{6} \pm \sqrt{6}}{2} = \frac{\sqrt{6}}{\sqrt{2}} \frac{(1 \pm i)}{\sqrt{2}} = \sqrt{3} e^{\pm i \frac{\pi}{4}}$$

$$\alpha = \sqrt{3} e^{i \frac{\pi}{4}}; \frac{\pi}{\beta} = i \alpha \beta = 3$$

$$\frac{\alpha^{99}}{\beta} + \beta^{98} = \alpha^{98} i + \beta^{98}$$

$$= 3^{49} [i \cdot i + (-j)] 3^{49} (-1 - j)$$

$$n = 49, a = -1, B = -1$$

$$\text{Other method } \frac{\alpha^{99} + \beta^{99}}{\beta} = \frac{\alpha^{100}}{3} + \beta^{98} = 3^{49} (-1) + 3^{49} (-i)$$

**Question:** Distance of (2,4) from the line  $2x + y + 2 = 0$  measured parallel to the line  $\sqrt{3}x + y + 2 = 0$ :

**Solution:**

$$\left( 2 - \frac{\sqrt{3}r}{2}, 4 + \frac{r}{2} \right)$$

$$2 \left( 2 - \frac{\sqrt{3}r}{2} \right) + 4 + \frac{r}{2} + 2 = 0$$

$$10 - \sqrt{3}r + \frac{r}{2} = 0$$

$$10 = \left( \sqrt{3} - \frac{1}{2} \right) r$$

$$r = \frac{20}{2\sqrt{3} - 1}$$

**Question:**  $a_1, a_2, \dots$  are in G.P. such that  $a_1 = \frac{1}{8}, a_1 \neq a_2$

$$= \frac{1}{8}, a_1 \neq a_2$$

and every term is equal to arithmetic mean of its two successive terms. Find  $S_{20} - S_{18}$ .

**Solution:**

$$a_1 = \frac{a_2 + a_3}{2}$$

$$a = \frac{ar + ar^2}{2}$$

$$r^2 + r = 2$$

$$r = -2$$

$$S_{20} - S_{18}$$

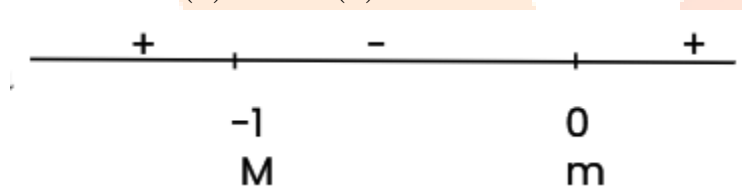
$$\frac{a[r^{20} - 1] - a[r^{10} - 1]}{r - 1}$$

$$\frac{a}{r - 1} [r^{20} - r^{18}]$$

$$\frac{1}{8(-3)} [2^{20} - 2^{18}]$$

$$\frac{2^{15}}{-3} [3] = -2^{15}$$

**Question:**  $f(x) = 2x + 3(x)^{\frac{2}{3}}, x \in \mathbb{R}$ .



**Options:**

- (a) it has one maxima no minima
- (b) it has one minima no maxima
- (c) it has 2 maxima and 1 minima
- (d) it has 1 maxima and 1 minima

**Solution:**

$$f(x) = 2x + 3(x)^{\frac{2}{3}}$$

$$f'(x) = 2 + 2x^{-1/3} = 0 = 2 \frac{(x^{1/3} + 1)}{x^{1/3}}$$

$$x = -1$$

$$x = -1 \text{ Maxima}$$

$$x = 0 \text{ Minima}$$

**Question:**  $R : \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$  if  $(1, 4), (1, 2) \in R$

Find the number of possible equivalence relations.

**Solution:**  $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 4), (4, 1), (1, 2), (2, 1), (4, 2), (2, 4)\}$

Other  $\{(1, 2), (3, 1), (2, 3), (3, 2), (4, 3), (3, 4)\}$

Either they all will come on not so 2.

**Question:** There are 8 identical book and 4 identical shelves Find the number of ways to arrange the book such that any shelf may be empty, every shelf can accommodate all the books

**Options:**

- (a) 13
- (b) 14
- (c) 15
- (d) 16

**Solution:**

|                |                |                |
|----------------|----------------|----------------|
| $(8, 0, 0, 0)$ | $(5, 3, 0, 0)$ | $(4, 3, 1, 0)$ |
| $(7, 1, 0, 0)$ | $(5, 2, 1, 0)$ | $(4, 2, 2, 0)$ |
| $(6, 2, 0, 0)$ | $(5, 2, 1, 0)$ | $(4, 2, 2, 0)$ |
| $(6, 1, 1, 0)$ | $(4, 4, 0, 0)$ | $(3, 3, 1, 1)$ |
| $(2, 2, 2, 2)$ | $(3, 3, 2, 0)$ | $(3, 2, 2, 1)$ |

Total 15 ways.

**Question:** Find the probability that number selected from 1 to 50 such that number is divisible by at least one out of 4, 6 or by 7

**Options:**

- (a)  $\frac{21}{25}$
- (b)  $\frac{18}{50}$
- (c)  $\frac{8}{25}$
- (d)  $\frac{21}{25}$

**Solution:**  $n(A \cup B \cup C)$

$$= 12 + 8 + 7 - 4 - 1 - 1 + 0$$

$$= 21$$

$$\text{Probability} = \frac{21}{25}$$

**Question:** A.B point of  $L_1$ , &  $L_2$  of shortest distance points from origin OA.OB